Algorithm: Have N^2 updates - "greedy" algorithm

Use knowledge that ~80% of spins align with ext. field

Then figure out which spins align with field and which with other spins - get approximation of results

Sort by external magnetic field

Should be better than ~6% above ground state

Sherrington - Kirkpatrick type matrix

Row sum calculate → make that the external field

Adjust variables to minimise energy

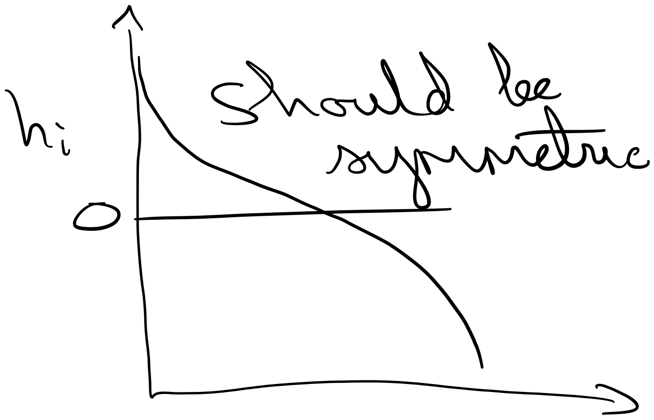
Bond matrix $J_{ij} \sim \mathcal{N}(0,1)$

Treat as QUBO problem: $x_i \in \{0,1\}$

$$H = -\sum_{ij} J_{ij}\sigma_i\sigma_j - \sum_i \sigma_i h_i$$

where $h_i = \sum_j J_{ij}$

Find ground ~~state~~ - exploit that spins should maximally align



- rank ordering of external fields

$P(h_i)$ = Gaussian

$$[Z^n] = \mathrm{Tr}\left\{\prod_{i<j}\int dx\, P(x)\exp(x g_{ij})\right\}$$

$$g_{ij} = \beta\sum_{\alpha=1}^{n}\left(\sigma_i^\alpha\sigma_j^\alpha + \sigma_i^\alpha + \sigma_j^\alpha\right)$$

$$P(x) = \frac{1}{J}\sqrt{\frac{N}{2\pi}}\exp\left(-\frac{N}{2J^2}x^2\right) \quad \text{set } J=1$$

$$\sqrt{\frac{N}{2\pi}}\int_{-\infty}^{\infty} dx\, e^{\underbrace{-N/2\, x^2 + x g_{ij}}}$$

$$= -\frac{N}{2}\left(x - \left(\frac{g_{ij}}{N}\right)\right)^2 - \left(\frac{g_{ij}}{N}\right)^2$$

$$= -\frac{N}{2}\left(x - \frac{g_{ij}}{N}\right)^2 + \frac{N}{2}\left(\frac{g_{ij}}{N}\right)^2$$

$$= \mathrm{Tr}_{n,N} \left\{ \prod_{i<j} \prod e^{N/2 \left(\frac{g_{ij}}{N}\right)^2} \right\}$$

Instead of introducing an integral over a delta function $\int dy\, \delta(y - g_{ij})$

we can do a different transform

$$g_{ij}^2 = \beta^2 \sum_{\alpha=1}^{n} \sum_{\gamma=1}^{n} \sigma_i^\alpha \sigma_j^\alpha \sigma_i^\gamma \sigma_j^\gamma +$$

$$\sigma_i^\alpha \sigma_j^\alpha \sigma_i^\gamma + \sigma_i^\alpha \sigma_j^\alpha \sigma_j^\gamma +$$

$$\sigma_i^\alpha \sigma_j^\gamma + \sigma_i^\alpha \sigma_i^\gamma + \sigma_j^\alpha \sigma_j^\gamma$$

$$+ \sigma_i^\alpha \sigma_i^\gamma \sigma_j^\gamma + \sigma_j^\alpha \sigma_i^\gamma \sigma_j^\gamma$$

$$+ \sigma_j^\alpha \sigma_i^\gamma$$

So we get

$$\frac{1}{2N} \sum_{i<j}\sum g_{ij}^2 = \frac{\beta^2}{2N} \sum_\alpha \sum_\gamma \underbrace{\{ - '' - \}_{ij}}_{= \frac{1}{2} \underset{\substack{\uparrow \\ \text{unrestricted}}}{\sum_i} \sum_j \{ \}_{ij} - \sum_i \{ \}_{ii}}$$

$$= \frac{\beta^2}{2N} \sum_\alpha \sum_\gamma \Bigg(\frac{1}{2} \Big(\sum_i \sigma_i^\alpha \sigma_i^\gamma \sum_j \sigma_j^\alpha \sigma_j^\gamma$$

$$+ \sum_i \sigma_i^\alpha \sigma_i^\gamma \sum_j \sigma_j^\alpha + \sum_i \sigma_i^\alpha \sum_j \sigma_j^\alpha \sigma_j^\gamma$$

$$+ \sum_i \sigma_i^\alpha \sum_j \sigma_j^\gamma + N \sum_i \sigma_i^\alpha \sigma_i^\gamma +$$

$$N \sum_j \sigma_j^\alpha \sigma_j^\gamma + \sum_i \sigma_i^\alpha \sigma_i^\gamma \sum_j \sigma_j^\gamma$$

$$+ \sum_i \sigma_i^\gamma \sum_j \sigma_j^\alpha \sigma_j^\gamma + \sum_i \sigma_i^\gamma \sum_j \sigma_j^\alpha)$$

$$- \Big(N + 2 \sum_i \sigma_i^\gamma + 4 \sum_i \sigma_i^\alpha \sigma_i^\gamma$$

$$+ 2 \sum_i \sigma_i^\alpha \Big) \Big)$$

Can combine terms that only differ by dummy indices:

$$[Z^n] = \mathrm{Tr}_{n,N} \Big\{ \exp \Big\{ \frac{\beta^2}{2N} \sum_\alpha \sum_\beta \Big[\frac{1}{2} \Big(\sum_i \sigma_i^\alpha \sigma_i^\gamma \Big)^2$$

$$+ \Big(\sum_i \sigma_i^\alpha \Big) \Big(\sum_i \sigma_i^\alpha \sigma_i^\gamma \Big)$$

$$+ \Big(\sum_i \sigma_i^\gamma \Big) \Big(\sum_i \sigma_i^\alpha \sigma_i^\gamma \Big)$$

$$+ N \Big(\sum_i \sigma_i^\alpha \sigma_i^\gamma \Big) + \Big(\sum_i \sigma_i^\alpha \Big) \Big(\sum_i \sigma_i^\gamma \Big)$$

$$- N - 2 \sum_i \sigma_i^\gamma - 4 \sum_i \sigma_i^\alpha \sigma_i^\gamma$$

$$- 2 \sum_i \sigma_i^\alpha \Big] \Big\} \Big\}$$